Ergodic Secrecy Rate of Movable Antenna-Aided MIMOME Systems with Statistical ECSI

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***Abstract*—This paper investigates the potential of movable antennas (MAs) to enhance physical layer security within a multiple-input multiple-output multiple-antenna eavesdropper (MIMOME) system. In this paper, we consider a practical scenario involving imperfect eavesdropper channel state information (ECSI) at the transmitter. Specifically, the ECSI incorporates an instantaneous line-of-sight (LoS) component and statistical non-line-of-sight (NLoS) components. To rigorously quantify secrecy performance under this inherent uncertainty, we adopt the ergodic secrecy rate (ESR) as the primary metric. Given the analytical intractability of the exact ESR, we employ random matrix theory to derive a deterministic equivalent. This approach can circumvent the need for extensive Monte Carlo simulations and reveal the explicit relationship between channel spatial statistics and secrecy performance. Based on these theoretical insights, we formulate an optimization problem for continuous antenna positioning. Simulation results validate our derivation and demonstrate that the proposed MA-aided scheme significantly outperforms conventional fixed-position antenna systems in terms of ESR.**

***Index Terms*—Ergodic secrecy rate, movable antenna system, physical layer security, random matrix theory.**

I. Introduction

The advent of sixth-generation (6G) wireless networks promises to support high-demand applications such as autonomous vehicles and the low-altitude economy [2]–[4]. However, traditional fixed-position antenna (FPA) systems struggle to provide the necessary ubiquitous coverage due to their limited spatial degrees of freedom [5]. To overcome this bottleneck, movable antenna (MA) systems have emerged as a promising solution, enabling transmitters to dynamically adjust antenna elements within a constrained region [6]–[9]. This capability allows for the proactive reshaping of channel responses, significantly enhancing system capacity compared to static arrays. Concurrently, the open broadcast nature of 6G heightens security risks, making physical-layer security (PLS) essential. While MA systems offer new geometric opportunities to bolster secrecy by enhancing legitimate channels and nulling eavesdroppers, achieving secure beamforming remains challenging. In practice, acquiring precise eavesdropper channel state information (ECSI) from passive, non-cooperative eavesdroppers is rarely feasible.

However, existing research into MA-enabled PLS often relies on overly idealistic assumptions, such as full knowledge of instantaneous ECSI [10], or utilizes simplified models that overlook stochastic multipath effects and multiple-antenna eavesdroppers (MIMOME) [11]. Current robust designs either ignore non-line-of-sight (NLoS) components or fail to address the complexities of matrix-valued channels inherent in scenarios where eavesdroppers also possess multiple antennas [12]. Consequently, there remains a critical gap in evaluating and optimizing secrecy performance in realistic environments where only statistical information regarding the eavesdropper's channel is available.

This paper explores the security benefits of MAs in a MIMOME configuration. Unlike traditional setups, our framework operates under statistical ECSI, incorporating both line-of-sight (LoS) and NLoS propagation to reflect realistic environments. By leveraging Random Matrix Theory (RMT), we derive an analytical deterministic equivalent for the Ergodic Secrecy Rate (ESR), obviating the need for time-consuming Monte Carlo simulations. To harness the spatial gains of MAS, we tackle the complex continuous position optimization problem through a tailored AMSGrad-based algorithm. This gradient-driven method ensures efficient and stable optimization of the antenna array geometry. Our findings demonstrate that MAs integration can provide a significant performance leap in PLS, which offers a robust defense against advanced eavesdropping threats.

II. System Model

This paper investigates a downlink secure MIMOME transmission scenario involving a base station (Alice) and a legitimate receiver (Bob).The BS is equipped with an array of N MAs located in a two-dimensional planar region. Unlike conventional FPA arrays, the MAs possess the capability to move continuously within a shared rectangular area, thereby exploiting additional spatial degrees of freedom. Specifically, each antenna element is constrained to move within a shared rectangular region of dimensions $D_x \times D_y$, where D_x and D_y denote the maximum movement ranges along the x-axis and y-axis, respectively. Mathematically, we have $-\mathbf{t}_{\max} \leq \mathbf{t}_n \leq \mathbf{t}_{\max}$, where $\mathbf{t}_{\max} = [D_x, D_y]^{\mathrm{T}}$. Bob is equipped with uniform linear array with L antennas. Alice transmits M data stream to Bob ($M \leq N, L$). The eavesdropper (Eve) equips M FPAs.

An extended version of this paper is available on arXiv [1].

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A. Channel Model

Given the covert nature of passive eavesdropping, the eavesdropper is typically modeled as Rician fading. In particular, the channel vectors between Alice to the mth antenna of Eve is denoted by [13]

$$\mathbf{g}_m = \sqrt{\frac{K_{e,m}\beta_{e,m}}{K_{e,m}+1}}\mathbf{g}_m^{\mathrm{LoS}}(\mathbf{T}) + \sqrt{\frac{\beta_{e,m}}{K_{e,m}+1}}\mathbf{g}_m^{\mathrm{NLoS}}, \quad (1)$$

where $K_{e,m}$ and $\beta_{e,m}$ denote the Rician K-factor and the path loss corresponding to the mth antenna of Eve. In particular, the LoS deterministic component is given by

$$\mathbf{g}_m^{\mathrm{LoS}}(\mathbf{T}) = \mathbf{a}(\mathbf{T}, \boldsymbol{\rho}_{e,m}), \quad (2)$$

where $\mathbf{a}(\mathbf{T}, \boldsymbol{\rho}_{e,m})$ denotes the field-response vector, i.e.,

$$\mathbf{a}(\mathbf{T}, \boldsymbol{\rho}_{e,m}) = \left[e^{j\frac{2\pi}{\lambda}\mathbf{t}_1^{\mathrm{T}}\boldsymbol{\rho}_{e,m}}, e^{j\frac{2\pi}{\lambda}\mathbf{t}_2^{\mathrm{T}}\boldsymbol{\rho}_{e,m}}, \cdots, e^{j\frac{2\pi}{\lambda}\mathbf{t}_N^{\mathrm{T}}\boldsymbol{\rho}_{e,m}}\right]^{\mathrm{T}}. \quad (3)$$

In (3), $\boldsymbol{\rho}_{e,m} = \left[\sin\theta_{e,m}\cos\varphi_{e,m}, \cos\theta_{e,m}\right]^{\mathrm{T}}$, where $\theta_{e,m}$ and $\varphi_{e,m}$ are the elevation and azimuth angles of the LoS path corresponding to the mth antenna of Eve, respectively. The wavelength is denoted by λ. The NLoS component $\mathbf{g}_m^{\mathrm{NLoS}}(\mathbf{T})$ consists of i.i.d. small-scale fading entries, each distributed as $\mathcal{CN}(0,1)$.

Unlike passive eavesdroppers, Bob is an authorized communication partner and therefore does not require covert transmission. The channel between Alice and Bob is assumed to be perfectly known, which is given by [7]

$$\mathbf{H} = \mathbf{A}_T\boldsymbol{\Lambda}_b\mathbf{A}_R^{\mathrm{H}}, \quad (4)$$

where $\mathbf{A}_T = \left[\mathbf{a}(\mathbf{T}, \boldsymbol{\rho}_{b,1}), \mathbf{a}(\mathbf{T}, \boldsymbol{\rho}_{b,2}), \cdots, \mathbf{a}(\mathbf{T}, \boldsymbol{\rho}_{b,L})\right]$, $\boldsymbol{\Lambda}_b = \mathrm{diag}\left(\beta_{b,l}, l = 1, 2, \cdots, L\right)$, and $\mathbf{A}_R = [\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_L]$. In particular, $\beta_{b,l}$ denote the path loss corresponding to the lth path, $\boldsymbol{\rho}_{b,l} = \left[\sin\theta_{b,l}\cos\varphi_{b,l}, \cos\theta_{b,l}\right]^{\mathrm{T}}$, where $\theta_{b,l}$ and $\varphi_{b,l}$ are the elevation and azimuth angles of the lth path at Alice, and $\mathbf{b}_l$ denotes the field-response vector of the lth path.

B. Received Signal Model

The received signal at Bob is given by

$$\mathbf{y}_b = \mathbf{H}^{\mathrm{H}}\mathbf{F}\mathbf{s} + \mathbf{n}_b, \quad (5)$$

where $\mathbf{F} \in \mathbb{C}^{N\times M}$ denotes the precoding vector, $\mathbf{s}$ denotes the transmitted signal, and $\mathbf{n}_b$ denotes the additive white Gaussian noise with zero mean and covariance $\sigma^2\mathbf{I}$. The transmitted symbol $\mathbf{s}$ are independent and identically distributed (i.i.d.) drawn from a standard complex constellation [14], whose kth entry s_k satisfies $\mathbb{E}[|s_k|^2] = 1$, $\mathbb{E}[s_k] = 0$, $\mathbb{E}[s_k^2] = 0$.

Since the eavesdroppers can cooperate with each other, they effectively form a multi-antenna receiver. The received signal at the effective multi-antenna eavesdropper is given by

$$\mathbf{y}_e = \mathbf{G}^{\mathrm{H}}\mathbf{F}\mathbf{s} + \mathbf{n}_e, \quad (6)$$

where $\mathbf{G} = [\mathbf{g}_1, \mathbf{g}_2, \cdots, \mathbf{g}_M] \in \mathbb{C}^{N\times M}$ denotes the eavesdropper channel, and $\mathbf{n}_e$ denotes the additive white Gaussian noise with zero mean and covariance $\sigma^2\mathbf{I}$. Specifically, an eavesdropper with M antennas can be viewed as a special case of our model where M single-antenna users are co-located.

III. Ergodic Secrecy Rate

Typically, secrecy performance is quantified by the secrecy rate, which characterizes the information rate that can be delivered reliably to the legitimate receiver while remaining confidential from Eves. In particular, the secrecy rate to Bob is given by [15]

$$\mathcal{R}(\mathbf{T}) = \left[\mathcal{R}_b(\mathbf{T}) - \mathcal{R}_e(\mathbf{T})\right]^+. \quad (7)$$

where

$$\begin{aligned} \mathcal{R}_b(\mathbf{T}) &= \log\left|\mathbf{I} + \sigma^{-2}\mathbf{H}^{\mathrm{H}}\mathbf{F}\mathbf{F}^{\mathrm{H}}\mathbf{H}\right|, \\ \mathcal{R}_e(\mathbf{T}) &= \log\left|\mathbf{I} + \sigma^{-2}\mathbf{G}^{\mathrm{H}}\mathbf{F}\mathbf{F}^{\mathrm{H}}\mathbf{G}\right|. \end{aligned} \quad (8)$$

A. ESR with statistic ECSI

Owing to the unavailability of instantaneous CSI for the eavesdroppers, the ESR is adopted as the performance metric. Specifically, the ESR achievable at Bob is defined as

$$\overline{\mathcal{R}}(\mathbf{T}) = \left[\mathcal{R}_b(\mathbf{T}) - \overline{\mathcal{R}}_e(\mathbf{T})\right]^+, \quad (9)$$

where the average rate to the eavesdroppers is given by

$$\overline{\mathcal{R}}_e(\mathbf{T}) = \mathbb{E}_{\mathbf{G}}\left[\mathcal{R}_e(\mathbf{T})\right]. \quad (10)$$

In particular, the expectation $\mathbb{E}_{\mathbf{G}}[\cdot]$ is taken over the small-scale fading distribution of the eavesdropping channels.

B. Asymptotic approximation of ESR

To derive a tractable expression for the ESR under the statistical model for the large-scale gains of eavesdroppers, we state the following proposition.

Proposition 1: Define the matrix $\mathbf{B}$ as

$$\mathbf{B} = \boldsymbol{\Lambda}^{\frac{1}{2}}\mathbf{G}_{\mathrm{LoS}}^{\mathrm{H}}\mathbf{F}. \quad (11)$$

where $\mathbf{G}_{\mathrm{LoS}} = \left[\mathbf{g}_1^{\mathrm{LoS}}(\mathbf{T}), \mathbf{g}_2^{\mathrm{LoS}}(\mathbf{T}), \cdots, \mathbf{g}_M^{\mathrm{LoS}}(\mathbf{T})\right]$, and

$$\boldsymbol{\Lambda} = \mathrm{diag}\left(\frac{K_{e,m}\beta_{e,m}}{K_{e,m}+1}, m = 1, 2, \cdots, M\right). \quad (12)$$

Define two diagonal matrices $\mathbf{D}$ and $\widetilde{\mathbf{D}}$ as

$$\mathbf{D} = \mathrm{diag}\left(\frac{M\beta_{e,m}}{K_{e,m}+1}, m = 1, 2, \cdots, M\right), \quad (13)$$

$$\widetilde{\mathbf{D}} = \mathrm{diag}\left(||\mathbf{f}_n||^2, n = 1, 2, \cdots, M\right). \quad (14)$$

Then, define $(\delta, \tilde{\delta})$ as the unique positive solution of the following fixed-point equation system:

$$\begin{cases} \delta = \frac{1}{M}\mathrm{tr}\left(\mathbf{D}\boldsymbol{\Gamma}\right) \\ \tilde{\delta} = \frac{1}{M}\mathrm{tr}\left(\widetilde{\mathbf{D}}\widetilde{\boldsymbol{\Gamma}}\right) \end{cases}, \quad (15)$$

where

$$\begin{cases} \boldsymbol{\Gamma} = \left[\sigma^2\boldsymbol{\Phi}^{-1} + \mathbf{B}\widetilde{\boldsymbol{\Phi}}\mathbf{B}^{\mathrm{H}}\right]^{-1} \\ \widetilde{\boldsymbol{\Gamma}} = \left[\sigma^2\widetilde{\boldsymbol{\Phi}}^{-1} + \mathbf{B}^{\mathrm{H}}\boldsymbol{\Phi}\mathbf{B}\right]^{-1} \\ \boldsymbol{\Phi} = \left(\mathbf{I} + \tilde{\delta}\mathbf{D}\right)^{-1} \\ \widetilde{\boldsymbol{\Phi}} = \left(\mathbf{I} + \delta\widetilde{\mathbf{D}}\right)^{-1} \end{cases}. \quad (16)$$

As $N, M \to \infty$, the ergodic secrecy rate can be given by

$$\overline{\mathcal{R}}_e \xrightarrow{a.s.} -\log\left|\sigma^2\boldsymbol{\Gamma}\right| + \log\left|\mathbf{I} + \delta\widetilde{\mathbf{D}}\right| - \sigma^2 M\delta\tilde{\delta}. \tag{17}$$

Proof: The main idea of this proof is to recast $\mathcal{R}_e$ into an extended model which fits into the framework of [16, Theorem 1]. First, we have

$$\mathcal{R}_e = \log\left|\mathbf{I} + \frac{1}{\sigma^2}\mathbf{Z}\mathbf{Z}^{\mathrm{H}}\right|, \tag{18}$$

where $\mathbf{Z} = \mathbf{G}^{\mathrm{H}}\mathbf{F}$. Note that the (m,n)th entry of $\mathbf{Z}$ is Gaussian, with mean $[\mathbf{B}]_{m,n}$ and variance given by $\sigma^2_{z,(m,n)} = \frac{\beta_{e,m}\|\mathbf{f}_n\|^2}{K_{e,m}+1}$, where $\mathbf{f}_n$ denotes the nth column of $\mathbf{F}$. Let $\mathbf{X}$ be a matrix whose entries are i.i.d. with mean 0 and variance 1. Then, $\mathbf{Z}$ is equivalent to

$$\mathbf{Z} \sim \mathbf{B} + \frac{1}{\sqrt{M}}\mathbf{D}^{\frac{1}{2}}\mathbf{X}\widetilde{\mathbf{D}}^{\frac{1}{2}}, \tag{19}$$

where $\mathbf{B}$, $\mathbf{D}$, and $\widetilde{\mathbf{D}}$ are defined in (11), (13), and (14), respectively. Based on [16, Theorem 1], we have

$$\mathbb{E}_{\mathbf{G}}(\mathcal{R}_e) \xrightarrow{a.s.} -\log\left|\sigma^2\boldsymbol{\Gamma}\right| + \log\left|\mathbf{I} + \delta\widetilde{\mathbf{D}}\right| - \sigma^2 M\delta\tilde{\delta}, \tag{20}$$

where $(\delta, \tilde{\delta})$ is defined as the solution to the fixed-point equation in (15). ■

IV. Antenna Position Optimization

In this paper, we propose to optimize the antenna position $\mathbf{T}$ to maximize the ESR. Concretely, the optimization problem is formulated as

$$\begin{aligned} \mathcal{P}_0 : \max_{\mathbf{T}} \quad & \mathcal{R}_b(\mathbf{T}) - \overline{\mathcal{R}}_e(\mathbf{T}) \\ s.t. \quad & (\mathrm{C1}) : \|\mathbf{t}_n - \mathbf{t}_m\|^2 \geq d, \quad m \neq n, \\ & (\mathrm{C2}) : \mathbf{t}_n \in \mathcal{T}, \quad n = 1, 2, \cdots, N, \end{aligned} \tag{21}$$

where C1 requires that the distance between any two distinct antennas must be at least the prescribed constant d, and C2 restricts each antenna to lie within a prescribed region $\mathcal{T} = \{\mathbf{t}_n | -\mathbf{t}_{\max} \leq \mathbf{t}_n \leq \mathbf{t}_{\max}\}$.

To solve this problem, we propose to leverage the AMSGrad optimization algorithm to update $\mathbf{t}_n$. Next, we provide the gradient of $\overline{\mathcal{R}}_e(\mathbf{T})$ with respect to $\mathbf{t}_n$. By differentiating the equations in (15) w.r.t. the ith entry of $\mathbf{t}_n$, denoted by $t_{n,i}$, we establish the following fixed-point equations.

$$\begin{cases} \delta'_{(n,i)} = \frac{1}{M}\mathrm{tr}\left(\mathbf{D}\boldsymbol{\Gamma}'_{(n,i)}\right) \\ \tilde{\delta}'_{(n,i)} = \frac{1}{M}\mathrm{tr}\left(\widetilde{\mathbf{D}}\widetilde{\boldsymbol{\Gamma}}'_{(n,i)}\right) \end{cases}, \tag{22}$$

where

$$\boldsymbol{\Gamma}'_{(n,i)} = -\boldsymbol{\Gamma}\left(\sigma^2\tilde{\delta}'_{(n,i)}\mathbf{D} + \boldsymbol{\Xi}'_{(n,i)}\right)\boldsymbol{\Gamma}, \tag{23a}$$

$$\widetilde{\boldsymbol{\Gamma}}'_{(n,i)} = -\widetilde{\boldsymbol{\Gamma}}\left(\sigma^2\delta'_{(n,i)}\widetilde{\mathbf{D}} + \widetilde{\boldsymbol{\Xi}}'_{(n,i)}\right)\widetilde{\boldsymbol{\Gamma}}, \tag{23b}$$

$$\boldsymbol{\Phi}'_{(n,i)} = -\boldsymbol{\Phi}\left(\tilde{\delta}'_{(n,i)}\mathbf{D}\right)\boldsymbol{\Phi}, \tag{23c}$$

$$\widetilde{\boldsymbol{\Phi}}'_{(n,i)} = -\widetilde{\boldsymbol{\Phi}}\left(\delta'_{(n,i)}\widetilde{\mathbf{D}}\right)\widetilde{\boldsymbol{\Phi}}, \tag{23d}$$

$$\boldsymbol{\Xi}'_{(n,i)} = \mathbf{B}\widetilde{\boldsymbol{\Phi}}'_{(n,i)}\mathbf{B}^{\mathrm{H}} + 2\Re\left(\boldsymbol{\Lambda}^{\frac{1}{2}}\tilde{\mathbf{g}}'_{n,i}\mathbf{e}_n^{\mathrm{T}}\mathbf{F}\widetilde{\boldsymbol{\Phi}}\mathbf{B}^{\mathrm{H}}\right), \tag{23e}$$

$$\widetilde{\boldsymbol{\Xi}}'_{(n,i)} = \mathbf{B}^{\mathrm{H}}\boldsymbol{\Phi}'_{(n,i)}\mathbf{B} + 2\Re\left(\mathbf{B}^{\mathrm{H}}\boldsymbol{\Phi}\boldsymbol{\Lambda}^{\frac{1}{2}}\tilde{\mathbf{g}}'_{n,i}\mathbf{e}_n^{\mathrm{T}}\mathbf{F}\right). \tag{23f}$$

The gradient of $\overline{\mathcal{R}}_e(\mathbf{T})$ w.r.t. $\mathbf{t}_n$ is denoted by $\boldsymbol{\nabla}_{e,\mathbf{t}_n} \triangleq \frac{\partial\overline{\mathcal{R}}(\mathbf{T})}{\partial\mathbf{t}_n}$, whose ith entry is given by

$$\begin{aligned} [\boldsymbol{\nabla}_{e,\mathbf{t}_n}]_i = & -\mathrm{tr}\left(\boldsymbol{\Gamma}^{-1}\boldsymbol{\Gamma}_{(n,i)}\right) + \tilde{\phi}\delta'_{(n,i)}\mathrm{tr}\left(\mathbf{F}\mathbf{F}^{\mathrm{H}}\right) \\ & - \sigma^2 M\left(\delta'_{(n,i)}\tilde{\delta} + \delta\tilde{\delta}'_{(n,i)}\right). \end{aligned} \tag{24}$$

Here, $t_{n,i}$ represents the ith entry of $\mathbf{t}_n$.

Given the gradient of $\overline{\mathcal{R}}_e(\mathbf{t}_n)$ is provided, we next derive the gradient of $\mathcal{R}_b(\mathbf{t}_n)$. To begin with, we introduce the steering-like vector $\mathbf{a}(\mathbf{t}_n)$ defined as $\mathbf{a}(\mathbf{t}_n) = \left[e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\boldsymbol{\rho}_{b,1}}, \ldots, e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\boldsymbol{\rho}_{b,L}}\right]^{\mathrm{T}} \in \mathbb{C}^{L\times 1}$. Then, the lth entry of $\mathbf{a}'_{n,i} = \frac{\partial\mathbf{a}(\mathbf{t}_n)}{\partial t_{n,i}}$ and $\mathbf{a}''_{n,(i,j)} = \frac{\partial^2\mathbf{a}(\mathbf{t}_n)}{\partial t_{n,i}\partial t_{n,j}}$ is respectively given by

$$\left[\mathbf{a}'_{n,i}\right]_l = j\frac{2\pi}{\lambda}[\boldsymbol{\rho}_{b,l}]_i e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\boldsymbol{\rho}_{b,l}}. \tag{25a}$$

$$\left[\mathbf{a}''_{n,(i,j)}\right]_l = -\frac{4\pi^2}{\lambda^2}[\boldsymbol{\rho}_{b,l}]_i[\boldsymbol{\rho}_{b,l}]_j e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\boldsymbol{\rho}_{b,l}}. \tag{25b}$$

Meanwhile, the gradient of $\mathcal{R}_b(\mathbf{t}_n)$ w.r.t. $\mathbf{t}_n$ is denoted by $\boldsymbol{\nabla}_{b,\mathbf{t}_n} \triangleq \frac{\partial\mathcal{R}_b(\mathbf{t}_n)}{\partial t_{n,i}}$, whose ith entry is given by

$$\left[\boldsymbol{\nabla}_{b,\mathbf{t}_n}\right]_i = \frac{2}{\sigma^2}\mathrm{tr}\left[\left(\mathbf{I} + \sigma^{-2}\mathbf{H}^{\mathrm{H}}\mathbf{F}_k\mathbf{F}_k^{\mathrm{H}}\mathbf{H}\right)^{-1}\Re\left(\mathbf{H}^{\mathrm{H}}\mathbf{F}_k\mathbf{F}_k^{\mathrm{H}}\mathbf{H}'_{n,i}\right)\right],$$

where $\mathbf{H}'_{n,i} = \mathbf{e}_n\left(\mathbf{a}'_{n,i}\right)^{\mathrm{T}}\boldsymbol{\Lambda}_b\mathbf{A}_R^{\mathrm{H}}$.

Note that the kth iteration for solving the problem (21) is given by

$$\mathbf{t}_{n,k+1} = \arg\min_{\mathbf{t}_n\in\mathcal{S}_T} -\mathcal{R}_b(\mathbf{t}_n;\mathbf{T}_k) + \overline{\mathcal{R}}_e(\mathbf{t}_n;\mathbf{T}_k), \tag{26}$$

whose gradient is given by $\boldsymbol{\nabla}_{\mathbf{t}_n} = -\boldsymbol{\nabla}_{b,\mathbf{t}_n} + \boldsymbol{\nabla}_{e,\mathbf{t}_n}$.

To solve this problem, we introduce another iteration. At the tth step of the inter-iteration, the first-order and second-order moments are defined by [17], [18]

$$\begin{aligned} \widehat{\mathbf{m}}_t &= \beta_{1,t}\widehat{\mathbf{m}}_{t-1} + (1-\beta_{1,t})\widehat{\mathbf{g}}_t, \\ \widehat{\mathbf{v}}_t &= \beta_2\widehat{\mathbf{v}}_{t-1} + (1-\beta_2)\left|\widehat{\mathbf{g}}_t\right|^2, \end{aligned} \tag{27}$$

where $\widehat{\mathbf{g}}_t = \boldsymbol{\nabla}_{\mathbf{t}_n}|_{\mathbf{t}_n=\mathbf{t}_{n,(k,t)}}$ denotes the gradient at the current iteration, and $\beta_2 \in (0,1)$. The intermediate parameter is set as $\beta_{1,t} = \beta_1\lambda_1^{t-1}$, where $\beta_1, \lambda_1 \in (0,1)$.

Then, the variable $\mathbf{t}_{n,(k,t+1)}$ can be updated by

$$\mathbf{t}_{n,(k,t+1)} = \min_{\mathbf{t}_n\in\mathcal{S}_T}\left\|\widehat{\mathbf{V}}_t^{\frac{1}{2}}\left(\mathbf{t}_n - \left(\mathbf{t}_{n,(k,t)} - \alpha_t\widehat{\mathbf{V}}_t^{-\frac{1}{2}}\widehat{\mathbf{m}}_t\right)\right)\right\|^2, \tag{28}$$

where $\widehat{\mathbf{V}}_t = \mathrm{diag}\left(\widehat{\mathbf{v}}_t\right)$, and the step size α_t is typically set to $\alpha/\sqrt{t}$ for some constant α. This problem is equivalent to a quadratic programming (QP) problem, i.e.,

$$\mathbf{t}_{n,(k,t+1)} = \min_{\mathbf{t}_n\in\mathcal{S}_T}\mathbf{t}_n^{\mathrm{T}}\widehat{\mathbf{V}}_t\mathbf{t}_n + 2\widehat{\mathbf{f}}_t^{\mathrm{T}}\mathbf{t}_n \tag{29}$$

where $\widehat{\mathbf{f}}_t = -\widehat{\mathbf{V}}_t\mathbf{t}_{n,(k,t)} + \alpha_t\widehat{\mathbf{V}}_t^{\frac{1}{2}}\widehat{\mathbf{m}}_t$. This problem (29) can be efficiently solved by using quadprog [19]. Once the convergence criterion is satisfied, the variable $\mathbf{t}_{n,k+1}$ is updated using $\mathbf{t}_{n,(k,t)}$. Subsequently, the precoder $\mathbf{F}_{k+1}$ is updated according to $\mathbf{T}_{k+1}$. In this paper, zero-forcing (ZF) precoding is adopted for simplicity.

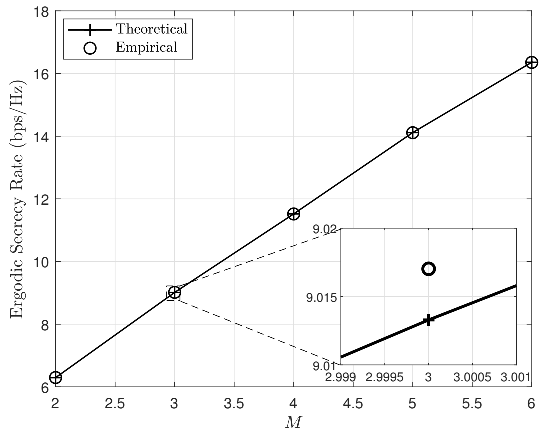

Fig. 1. The ergodic secrecy rate $\overline{\mathcal{R}}$ versus M.

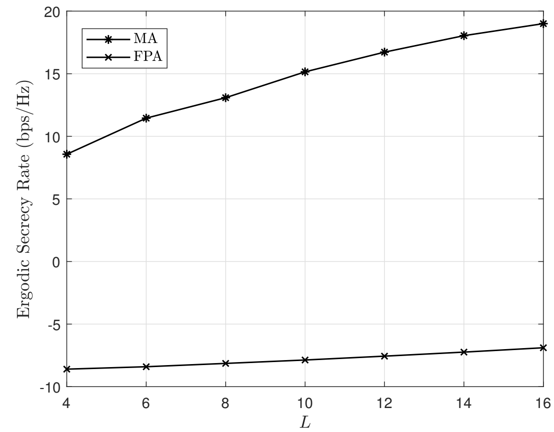

Fig. 2. The ergodic secrecy rate $\overline{\mathcal{R}}$ versus L.

V. Simulation Results

In this simulation, we will show the effectiveness of the proposed method. We consider a millimeter-wave (mmWave) system operating at a carrier frequency of 28 GHz. Unless specified otherwise, the transmitter and receiver are equipped with $N = 8$ and $M = 4$ antennas, respectively. The number of eavesdroppers is also set as $M = 4$. The feasible region of MA is set as $D_x = D_y = 50\lambda$. The K-Rician factors are set as $K_{e,1} = K_{e,2} = \cdots = K_{e,M} = K_e = 4$. The path gain for the l-th path in the legitimate channel and the mth Eve is modeled as $\beta_{b,l}, \beta_{e,m} \sim \mathcal{CN}(0, 10^{-0.1\vartheta(d)})$, where $\vartheta(d) = a + 10b\log_{10}(d) + \epsilon$, with d denoting the propagation distance and $\epsilon \sim \mathcal{CN}(0, \sigma_\epsilon^2)$ [20]. Following [20], we use $a = 61.4$, $b = 2$, and $\sigma_\epsilon = 5.8$ dB. The distance between the transmitter and receiver is set as 40 m. The number of pathes is set as $L = 16$. We consider one communication user and M eavesdropper whose AOD and AOA are randomly generated from $[10^\circ, 30^\circ]$ and $[40^\circ, 70^\circ]$, respectively. The noise power is set to $\sigma^2 = -90$ dBm. The parameters of AMSGrad-based method is set as $\beta_1 = 0.9$, $\lambda_1 = 0.99$, $\beta_2 = 0.9$, and $\alpha = 0.5$.

The legend "Empirical" represents the ESR performance obtained through Monte Carlo simulations, where each abscissa is averaged over 1000 independent trials. The legend "Theoretical" represents the deterministic equivalent for the ESR, which is defined in (17). As illustrated in Fig. 1, we investigate the accuracy of the deterministic approximation introduced in ***Proposition 1***. While the theoretical framework of ***Proposition 1*** is rigorously predicated on the asymptotic regime where both the number of antennas N and the number of eavesdroppers M approach infinity, our numerical evaluations reveal a robust applicability to finite-dimensional systems. Specifically, the approximation demonstrates high precision even for moderate-dimensional setting (e.g., $M \geq 3$), significantly relaxing the constraints typically associated with large-dimensional limits. Furthermore, the figure corroborates the law of RMT, as the approximation accuracy monotonically improves with an increasing number of snapshots. This trend confirms that the empirical results converge tightly to their deterministic equivalents as M grows.

Fig. 2 illustrates the impact of the number of paths, L, in Bob's channel on the ESR. This trend stems from the higher spatial DoF provided by larger antenna arrays, which facilitate more precise beamforming to enhance signal power at the legitimate receiver while suppressing leakage to the eavesdropper. Significantly, the proposed MA-based scheme consistently surpasses the baseline. In the context of PLS, a negative secrecy rate signifies that the eavesdropper channel surpasses that of the legitimate link, rendering the achievement of a non-zero secrecy capacity theoretically unfeasible. As illustrated in Fig. 2, the ESR of the traditional FPA scheme falls below 0 bps/Hz. In contrast, the MA scheme maintains a positive ESR by leveraging spatial degrees of freedom to actively reconfigure the channel, thereby ensuring secure transmission even in scenarios where the FPA systems fail.

VI. Conclusion

In this paper, we have investigated secure communication within a MIMOME system enhanced by MAs, specifically addressing the challenge of imperfect ECSI. We adopted a practical channel model assuming access to instantaneous LoS and statistical NLoS components, and employ the ESR as the performance metric. To overcome the analytical intractability of the exact ESR, we utilized large-dimensional RMT to derive a deterministic closed-form approximation. This analytical result not only facilitates efficient performance evaluation without Monte Carlo averaging but also provides the theoretical basis for system optimization. Guided by these derivations, we formulated a joint optimization problem to maximize the ESR by designing the MA positions. Simulation results have corroborated the accuracy of our RMT-based approximation and demonstrated that the proposed MA-aided secure transmission scheme significantly outperforms conventional FPA benchmarks, highlighting the potential of MAS in enhancing physical layer security.

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